

Answer Key — Solute, Solvent and Solution

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

- Q1** (b) **uniform mixture** — a solution is a uniform mixture.
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- Q2** (b) **solute** — sugar dissolves, so it is the solute.
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- Q3** (b) **larger amount** — the liquid in the larger amount is the solvent.
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- Q4** (c) **sugar** — sugar dissolves to give a uniform solution; the others don't.
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- Q5** (a) **Both true, R explains A** — water dissolves the sugar, so water is the solvent — as R explains.
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- Q6** **solution** — they combine to form a solution.
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- Q7** **solvent** — water is a good (universal) solvent.
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Short answer

- Q8** Solute = the substance that dissolves (salt); solvent = the substance it dissolves in (water); solution = the uniform mixture formed (salt water).
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- Q9** The liquid present in the smaller amount is the solute; the one in the larger amount is the solvent.
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- Q10** Salt dissolves evenly to give a uniform mixture (solution); sand stays separate, forming a non-uniform mixture.
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Long / application

- Q11** (i) When a solid dissolves in a liquid, the solid is the solute and the liquid the solvent (e.g. sugar = solute, water = solvent). (ii) When two liquids mix, the one in the smaller amount is the solute and the one in the larger amount is the solvent (e.g. a little acetic acid in much water → acid = solute, water = solvent).
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- Q12** The sugar and salt dissolve evenly throughout the water, so their particles are uniformly spread; every sip has the same amount of each. This shows ORS is a uniform mixture — a solution.
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